

**CISS362: Introduction to Automata Theory, Languages, and
Computation
Assignment a06**

Name: _____

OBJECTIVES

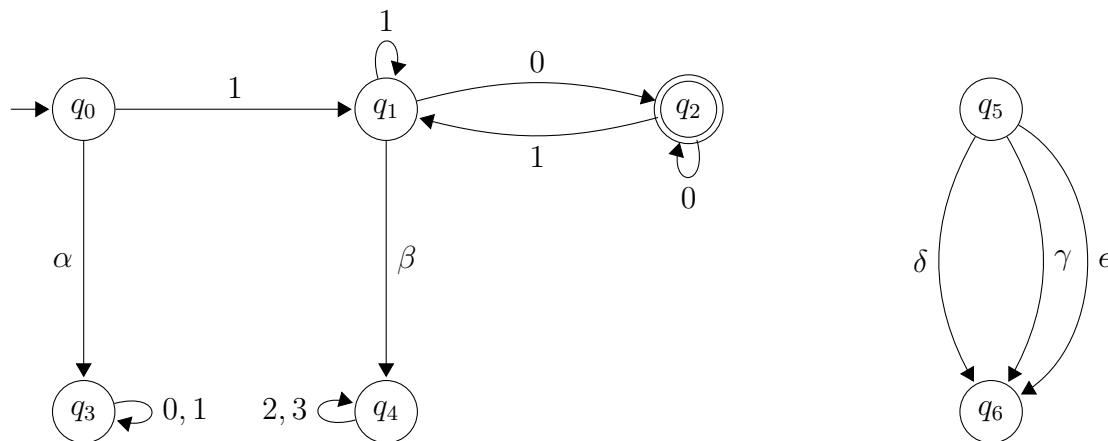
- Design DFAs
- Design NFAs.

As before modify the files `q01.tex` for Q1, `q02.tex` for Q2, etc. Tar and gzip your directory and email to `yliow.submit@gmail.com` using your college email account. Use subject line “`ciiss362 a06`”.

- Sipser 1.17. Q1
- Sipser 1.18. Q2. Solutions for (a)-(d) provided.
- Sipser 1.19. Q3
- Sipser 1.20. DIY. First solve it (use simplest examples). Then check the solution below.
- Sipser 1.21. Q4
- Sipser 1.22. Q5
- Sipser 1.23. Try it first. Then study the solution in the book.
- Sipser 1.28. Q6
- Sipser 1.31. Q7

HOW TO DRAW A STATE DIAGRAM

Here's an example showing you how to draw the elements of a state diagram. Also, look at the solution to 1.3 below.



For more information on drawing state diagrams go to my tutorials and look for `latex-automata.pdf`:

<https://drive.google.com/file/d/1AeE-POWNvQlitzPDxQpGE8bMR9Yc9gMW>

Let me know if you have any questions about drawing state diagram.

Q1. Sipser 1.17.

- (a) Design an NFA N that accepts the language $\{01, 001, 010\}^*$.
- (b) Convert N to an equivalent DFA using the powerset construction.

SOLUTION.

Q2. Sipser 1.18. Solutions for (a)-(d) provided.

Write down regular expressions generating the following languages where $\Sigma = \{0, 1\}$

- (a) $\{w \mid w \text{ begins with 1 and ends with 0}\}$
- (b) $\{w \mid w \text{ contains at least three 1s}\}$
- (c) $\{w \mid w \text{ contains substring 0101}\}$
- (d) $\{w \mid w \text{ has length at least 3 and its third symbol is a 0}\}$
- (e) $\{w \mid w \text{ start with 0 and has odd length, or starts with 1 and has even length}\}$
- (f) $\{w \mid w \text{ does not contain the substring 110}\}$
- (g) $\{w \mid \text{the length of } w \text{ is at most 5}\}$
- (h) $\{w \mid w \text{ is any string except 11 and 111}\}$
- (i) $\{w \mid \text{every odd position of } w \text{ is a 1}\}$
- (j) $\{w \mid w \text{ contains at least two 0s and at most one 1}\}$
- (k) $\{\epsilon, 0\}$
- (l) $\{w \mid w \text{ contains an even number of 0, or contains exactly two 1s}\}$
- (m) $\{\}$
- (n) $\{w \mid w \neq \epsilon\}$

SOLUTION.

- (a) Solution provided: $1(0 \cup 1)^*0$
- (b) Solution provided: $(0 \cup 1)^*1(0 \cup 1)^*1(0 \cup 1)^*1(0 \cup 1)^*$
- (c) Solution provided: $(0 \cup 1)^*0101(0 \cup 1)^*$
- (d) Solution provided: $(0 \cup 1)(0 \cup 1)0(0 \cup 1)^*$

Q3. Sipser 1.19.

SOLUTION.

Solution to Sipser 1.20:

- (a) Members: ϵ , ab . Nonmembers: ba , bab .
- (b) Members: ab , $abab$. Nonmembers: ϵ , a .
- (c) Members: ϵ , a . Nonmembers: ab , ba .
- (d) Members: ϵ , aaa . Nonmembers: a , b .
- (e) Members: aba , $aaba$. Nonmembers: ϵ , a .
- (f) Members: aba , bab . Nonmembers: ϵ , a .
- (g) Members: b , ab . Nonmembers: ϵ , a .
- (h) Members: a , ab . Nonmembers: ϵ , b .

Q4. Sipser 1.21.

SOLUTION.

Q5. Sipser 1.22.

SOLUTION.

Q6. Sipser 1.28.

SOLUTION.

Q7. Sipser 1.31.

Suppose $L = L(M)$ where $M = (\Sigma, Q, q_0, F, \delta)$.

- (a) Describe informally how to design a DFA that accepts L^R .
- (b) Describe formally a DFA, M^R , such that $L(M^R) = L^R$.

SOLUTION.

- (a)
- (b)
- (c)